Project Plan: Integrated Information Retrieval System with TF-IDF Ranking and Classification

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Project Scope:

1. **Design and implement an information retrieval system**: Indexing and searching a collection of documents.
2. **TF-IDF Ranking**: Assigning weights to terms based on frequency and importance.
3. **Classification Algorithms**: Using Naive Bayes or Support Vector Machines (SVM) for categorizing documents and queries.
4. **User Interface**: Developing an intuitive UI for querying the system and displaying results.
5. **Performance Evaluation**: Using precision, recall, and F1 score to measure system effectiveness.

Key Tasks:

1. **System Design:**

- Define architecture and components.
- Identify data structures for indexing (inverted index).
- Select retrieval algorithms (Boolean, Vector Space Model).
- Ensure scalability and efficiency.

2. **TF-IDF Implementation:**

- Calculate term frequency (TF).
- Calculate inverse document frequency (IDF).
- Compute TF-IDF scores for terms in documents.
- Rank documents based on TF-IDF scores for queries.

3. **Classification Integration:**

- Implement Naive Bayes or SVM for document classification.
- Train classifiers on labeled data.
- Categorize incoming queries and documents.

4. **Interface Design:**

- Develop a user-friendly interface (web-based or desktop).

- Include search functionality, display of ranked documents, and classified categories.
- Ensure usability and accessibility.

5. **System Implementation:**

- Use Python or Java for development.
- Implement indexing, TF-IDF calculation, and classification modules.
- Ensure efficient handling of large datasets.

6. **Evaluation:**

- Conduct experiments with and without TF-IDF ranking.
- Measure performance using precision, recall, and F1 score.
- Compare with baseline retrieval systems.